

# Operations Research

## Transportation Problem — Complete Solution

**Name:** Nadir Ali Khan  
**Roll No:** MT-0123-052  
**Subject:** Operations Research  
**Instructor:** Sir Darshan  
**Department:** BS Mathematics

### Problem Statement

Given the following transportation problem:

| Origin        | 1  | 2  | 3  | 4  | Supply |
|---------------|----|----|----|----|--------|
| A             | 6  | 9  | 7  | 15 | 50     |
| B             | 3  | 5  | 8  | 10 | 80     |
| C             | 15 | 6  | 4  | 7  | 40     |
| D             | 3  | 9  | 6  | 4  | 90     |
| <b>Demand</b> | 50 | 40 | 70 | 80 |        |

- (a) Find the initial feasible solution using **Vogel's Approximation Method**. Obtain an optimal solution using the **Stepping-Stone Method**.
- (b) Find the initial feasible solution using **Least Cost Method**. Obtain an optimal solution using the **Modified Distribution (MODI) Method**.

**Balance Check:** Total Supply =  $50 + 80 + 40 + 90 = 260$ , Total Demand =  $50 + 40 + 70 + 80 = 240$ .

Supply > Demand by **20**. Add a **dummy destination D5** with demand 20 and zero costs.

| Origin        | 1  | 2  | 3  | 4  | D5 | Supply |
|---------------|----|----|----|----|----|--------|
| A             | 6  | 9  | 7  | 15 | 0  | 50     |
| B             | 3  | 5  | 8  | 10 | 0  | 80     |
| C             | 15 | 6  | 4  | 7  | 0  | 40     |
| D             | 3  | 9  | 6  | 4  | 0  | 90     |
| <b>Demand</b> | 50 | 40 | 70 | 80 | 20 | 260    |

## Part (a): Vogel's Approximation Method + Stepping Stone Method

### VAM — Penalty Iterations

Compute the **row and column penalties** (2nd minimum – minimum cost). Allocate to the minimum-cost cell of the highest-penalty row/column.

#### Iteration 1

|                | 1  | 2 | 3 | 4  | D5 | Supply | Penalty  |
|----------------|----|---|---|----|----|--------|----------|
| A              | 6  | 9 | 7 | 15 | 0  | 50     | <b>6</b> |
| B              | 3  | 5 | 8 | 10 | 0  | 80     | 3        |
| C              | 15 | 6 | 4 | 7  | 0  | 40     | 4        |
| D              | 3  | 9 | 6 | 4  | 0  | 90     | 3        |
| <b>Penalty</b> | 0  | 1 | 2 | 3  | 0  |        |          |

Max penalty = **6** (Row A). Min cost in Row A = 0 (D5). Allocate  $x_{A,D5} = \min(50, 20) = 20$ . **D5 eliminated**. Remaining supply of A = 30.

#### Iteration 2 — D5 removed

|                | 1  | 2 | 3 | 4        | Supply | Penalty |
|----------------|----|---|---|----------|--------|---------|
| A              | 6  | 9 | 7 | 15       | 30     | 1       |
| B              | 3  | 5 | 8 | 10       | 80     | 2       |
| C              | 15 | 6 | 4 | 7        | 40     | 2       |
| D              | 3  | 9 | 6 | 4        | 90     | 1       |
| <b>Penalty</b> | 0  | 1 | 2 | <b>3</b> |        |         |

Max penalty = **3** (Col 4). Min cost in Col 4 = 4 (Row D). Allocate  $x_{D,4} = \min(90, 80) = 80$ . **Col 4 eliminated**. Remaining supply of D = 10.

## Iteration 3 — Col 4 removed

|         | 1  | 2 | 3 | Supply | Penalty  |
|---------|----|---|---|--------|----------|
| A       | 6  | 9 | 7 | 30     | 1        |
| B       | 3  | 5 | 8 | 80     | 2        |
| C       | 15 | 6 | 4 | 40     | 2        |
| D       | 3  | 9 | 6 | 10     | <b>3</b> |
| Penalty | 0  | 1 | 2 |        |          |

Max penalty = **3** (Row D). Min cost in Row D = 3 (Col 1). Allocate  $x_{D,1} = \min(10, 50) = 10$ . **Row D eliminated.** Remaining demand Col 1 = 40.

## Iteration 4 — Row D removed

|         | 1        | 2 | 3        | Supply | Penalty |
|---------|----------|---|----------|--------|---------|
| A       | 6        | 9 | 7        | 30     | 1       |
| B       | 3        | 5 | 8        | 80     | 2       |
| C       | 15       | 6 | 4        | 40     | 2       |
| Penalty | <b>3</b> | 1 | <b>3</b> |        |         |

Max penalty = **3** (Col 1 & Col 3 tie; choose Col 1). Min cost = 3 (Row B). Allocate  $x_{B,1} = \min(80, 40) = 40$ . **Col 1 eliminated.** Remaining supply of B = 40.

## Iterations 5–7

**Iter 5:** Max penalty 3 (Row B / Col 3 tie; choose Col 3). Min cost in Col 3 = 4 (Row C).  $x_{C,3} = \min(40, 70) = 40$ . Row C eliminated. Col 3 remaining = 30.

**Iter 6:** Max penalty 4 (Col 2). Min cost = 5 (Row B).  $x_{B,2} = \min(40, 40) = 40$ . Both Row B & Col 2 eliminated.

**Iter 7:** Only Row A, Col 3 remain.  $x_{A,3} = \min(30, 30) = 30$ . Done.

## VAM Initial Basic Feasible Solution

| Route                       | Units | Cost | Total |
|-----------------------------|-------|------|-------|
| $A \rightarrow D5$          | 20    | 0    | 0     |
| $A \rightarrow 3$           | 30    | 7    | 210   |
| $B \rightarrow 1$           | 40    | 3    | 120   |
| $B \rightarrow 2$           | 40    | 5    | 200   |
| $C \rightarrow 3$           | 40    | 4    | 160   |
| $D \rightarrow 1$           | 10    | 3    | 30    |
| $D \rightarrow 4$           | 80    | 4    | 320   |
| Initial Transportation Cost |       |      | 1040  |

Note: 7 allocations vs.  $m + n - 1 = 8$  required  $\Rightarrow$  **degenerate**. Add  $\varepsilon \approx 0$  at cell  $(C, 2)$  to complete the basis.

## Stepping Stone Method — Optimality Check

Compute dual variables  $u_i, v_j$  (with  $u_A = 0$ ) using  $c_{ij} = u_i + v_j$  for each basic cell:

$$\begin{array}{ll}
 (A, 3): & v_3 = 7 \\
 (C, 3): & u_C = -3 \\
 (B, 2): & u_B = -4 \\
 (D, 1): & u_D = -4
 \end{array}
 \quad
 \begin{array}{ll}
 (A, D5): & v_{D5} = 0 \\
 (C, 2) = \varepsilon: & v_2 = 9 \\
 (B, 1): & v_1 = 7 \\
 (D, 4): & v_4 = 8
 \end{array}$$

Improvement index  $\Delta_{ij} = c_{ij} - u_i - v_j$  for all non-basic cells:

| Cell     | $c_{ij}$ | $u_i + v_j$  | $\Delta_{ij}$ | Status                |
|----------|----------|--------------|---------------|-----------------------|
| $(A, 1)$ | 6        | $0 + 7 = 7$  | -1            | Enter (most negative) |
| $(A, 2)$ | 9        | $0 + 9 = 9$  | 0             | Alternative optimum   |
| $(A, 4)$ | 15       | $0 + 8 = 8$  | +7            |                       |
| $(B, 3)$ | 8        | $-4 + 7 = 3$ | +5            |                       |
| $(B, 4)$ | 10       | $-4 + 8 = 4$ | +6            |                       |
| $(C, 1)$ | 15       | $-3 + 7 = 4$ | +11           |                       |
| $(C, 4)$ | 7        | $-3 + 8 = 5$ | +2            |                       |
| $(D, 2)$ | 9        | $-4 + 9 = 5$ | +4            |                       |
| $(D, 3)$ | 6        | $-4 + 7 = 3$ | +3            |                       |

Cell  $(A, 1)$  has  $\Delta_{A,1} = -1$ . Trace the **stepping stone loop**:

$$\underbrace{(A, 1)}_{+} \rightarrow \underbrace{(A, 2)}_{-\varepsilon} \rightarrow \underbrace{(B, 2)}_{+} \rightarrow \underbrace{(B, 1)}_{-} \rightarrow (A, 1)$$

$\theta = \min\{\varepsilon, 40\} = \varepsilon$ . **Degenerate pivot**:  $(A, 1)$  enters at  $\varepsilon$ ,  $(C, 2)$  exits. Allocations numerically unchanged.

Re-evaluating with new basis ( $v_1 = 6, v_2 = 8, v_4 = 7; u_B = u_C = u_D = -3$ ):

| Cell     | $c_{ij}$ | $u_i + v_j$ | $\Delta_{ij}$ | Status |
|----------|----------|-------------|---------------|--------|
| $(A, 2)$ | 9        | 8           | +1            |        |
| $(A, 4)$ | 15       | 7           | +8            |        |
| $(B, 3)$ | 8        | 4           | +4            |        |
| $(B, 4)$ | 10       | 4           | +6            |        |
| $(C, 1)$ | 15       | 3           | +12           |        |
| $(C, 2)$ | 6        | 5           | +1            |        |
| $(C, 4)$ | 7        | 4           | +3            |        |
| $(D, 2)$ | 9        | 5           | +4            |        |
| $(D, 3)$ | 6        | 4           | +2            |        |

All  $\Delta_{ij} \geq 0 \rightarrow$  **Optimal**.

#### Optimal Solution

**Optimal Allocation (Part a):**

| Route               | Units | Cost | Total       |
|---------------------|-------|------|-------------|
| $A \rightarrow 3$   | 30    | 7    | 210         |
| $B \rightarrow 1$   | 40    | 3    | 120         |
| $B \rightarrow 2$   | 40    | 5    | 200         |
| $C \rightarrow 3$   | 40    | 4    | 160         |
| $D \rightarrow 1$   | 10    | 3    | 30          |
| $D \rightarrow 4$   | 80    | 4    | 320         |
| <b>Minimum Cost</b> |       |      | <b>1040</b> |

*Origin A holds 20 units at the dummy destination (unused supply).*

**Part (b): Least Cost Method + MODI Method**

### Least Cost Method — Allocations

Repeatedly assign to the **globally cheapest available cell**.

#### LCM Step-by-Step

**Step 1.** Min cost = 0 (D5 column). Allocate  $x_{A,D5} = \min(50, 20) = \mathbf{20}$ . D5 column removed. Remaining A = 30.

**Step 2.** Min cost = 3 at (B, 1) & (D, 1); choose (B, 1).  $x_{B,1} = \min(80, 50) = \mathbf{50}$ . Col 1 removed. Remaining B = 30.

**Step 3.** Min cost = 4 at (C, 3) & (D, 4); choose (C, 3).  $x_{C,3} = \min(40, 70) = \mathbf{40}$ . Row C removed. Col 3 remaining = 30.

**Step 4.** Min cost = 4 at (D, 4).  $x_{D,4} = \min(90, 80) = \mathbf{80}$ . Col 4 removed. Remaining D = 10.

**Step 5.** Min cost = 5 at (B, 2).  $x_{B,2} = \min(30, 40) = \mathbf{30}$ . Row B removed. Col 2 remaining = 10.

**Step 6.** Min cost = 6 at (D, 3).  $x_{D,3} = \min(10, 30) = \mathbf{10}$ . Row D removed. Col 3 remaining = 20.

**Step 7.** Min cost = 7 at (A, 3).  $x_{A,3} = \min(30, 20) = \mathbf{20}$ . Col 3 removed. Remaining A = 10.

**Step 8.** Only (A, 2) left.  $x_{A,2} = \mathbf{10}$ .

## LCM Initial Basic Feasible Solution

| Route              | Units | Cost | Total |
|--------------------|-------|------|-------|
| $A \rightarrow D5$ | 20    | 0    | 0     |
| $A \rightarrow 2$  | 10    | 9    | 90    |
| $A \rightarrow 3$  | 20    | 7    | 140   |
| $B \rightarrow 1$  | 50    | 3    | 150   |
| $B \rightarrow 2$  | 30    | 5    | 150   |
| $C \rightarrow 3$  | 40    | 4    | 160   |
| $D \rightarrow 3$  | 10    | 6    | 60    |
| $D \rightarrow 4$  | 80    | 4    | 320   |
| Initial Cost       |       |      | 1070  |

8 basic allocations =  $m + n - 1 = 8$ . Non-degenerate. ✓

## MODI Method — Iteration 1

Set  $u_A = 0$  and solve  $c_{ij} = u_i + v_j$  for each basic cell:

$$\begin{array}{ll}
 (A, D5): & v_{D5} = 0 \\
 (A, 2): & v_2 = 9 \\
 (A, 3): & v_3 = 7 \\
 (B, 2): & u_B = -4 \\
 (B, 1): & v_1 = 7 \\
 (C, 3): & u_C = -3 \\
 (D, 3): & u_D = -1 \\
 (D, 4): & v_4 = 5
 \end{array}$$

| Cell     | $c_{ij}$ | $u_i + v_j$ | $\Delta_{ij}$ | Status                |
|----------|----------|-------------|---------------|-----------------------|
| $(A, 1)$ | 6        | 7           | -1            |                       |
| $(A, 4)$ | 15       | 5           | +10           |                       |
| $(B, 3)$ | 8        | 3           | +5            |                       |
| $(B, 4)$ | 10       | 3           | +9            |                       |
| $(C, 1)$ | 15       | 4           | +11           |                       |
| $(C, 2)$ | 6        | 6           | 0             |                       |
| $(C, 4)$ | 7        | 4           | +5            |                       |
| $(D, 1)$ | 3        | 6           | -3            | Enter (most negative) |
| $(D, 2)$ | 9        | 8           | +1            |                       |

Enter  $(D, 1)$ . Stepping stone loop:

$$\underbrace{(D, 1)}_{+} \rightarrow \underbrace{(D, 3)}_{-} \rightarrow \underbrace{(A, 3)}_{+} \rightarrow \underbrace{(A, 2)}_{-} \rightarrow \underbrace{(B, 2)}_{+} \rightarrow \underbrace{(B, 1)}_{-} \rightarrow (D, 1)$$

$$\theta = \min\{x_{D,3}, x_{A,2}, x_{B,1}\} = \min\{10, 10, 50\} = \mathbf{10}$$

Tie between  $(D, 3)$  and  $(A, 2)$ . Remove  $(D, 3)$ ; retain  $(A, 2) = \varepsilon$ .

Updated allocations after  $\theta = 10$ :

$$x_{D,1} += 10, \quad x_{A,3} += 10, \quad x_{B,2} += 10, \quad x_{D,3} = 0 \text{ (out)}, \quad x_{A,2} = \varepsilon, \quad x_{B,1} -= 10$$

$$\text{Cost improvement} = \theta \times |\Delta_{D,1}| = 10 \times 3 = 30 \implies \text{New cost} = 1070 - 30 = \mathbf{1040}$$

### MODI Method — Iteration 2

New dual variables with  $(A, 2) = \varepsilon$  in basis:  $v_1 = 7, v_2 = 9, v_3 = 7, v_4 = 8; u_B = -4, u_C = -3, u_D = -4$ .

| Cell     | $c_{ij}$ | $u_i + v_j$ | $\Delta_{ij}$ | Status       |
|----------|----------|-------------|---------------|--------------|
| $(A, 1)$ | 6        | 7           | -1            | <b>Enter</b> |
| $(A, 4)$ | 15       | 8           | +7            |              |
| $(B, 3)$ | 8        | 3           | +5            |              |
| $(B, 4)$ | 10       | 4           | +6            |              |
| $(C, 1)$ | 15       | 4           | +11           |              |
| $(C, 2)$ | 6        | 6           | 0             |              |
| $(C, 4)$ | 7        | 5           | +2            |              |
| $(D, 2)$ | 9        | 5           | +4            |              |
| $(D, 3)$ | 6        | 3           | +3            |              |

Enter  $(A, 1)$ . Loop:  $(A, 1)^+ \rightarrow (A, 2)^- \rightarrow (B, 2)^+ \rightarrow (B, 1)^- \rightarrow (A, 1)$

$\theta = \min\{\varepsilon, 40\} = \varepsilon$ . **Degenerate pivot.**  $(A, 2)$  exits,  $(A, 1) = \varepsilon$  enters. Cost change  $\approx 0$ .

### MODI Method — Final Verification

New duals:  $v_1 = 6, v_2 = 8, v_3 = 7, v_4 = 7; u_B = u_C = u_D = -3$ .



| Cell   | $c_{ij}$ | $u_i + v_j$ | $\Delta_{ij}$ | Status |
|--------|----------|-------------|---------------|--------|
| (A, 2) | 9        | 8           | +1            |        |
| (A, 4) | 15       | 7           | +8            |        |
| (B, 3) | 8        | 4           | +4            |        |
| (B, 4) | 10       | 4           | +6            |        |
| (C, 1) | 15       | 3           | +12           |        |
| (C, 2) | 6        | 5           | +1            |        |
| (C, 4) | 7        | 4           | +3            |        |
| (D, 2) | 9        | 5           | +4            |        |
| (D, 3) | 6        | 4           | +2            |        |

All  $\Delta_{ij} \geq 0 \rightarrow$  **Optimal solution reached.**

#### Optimal Solution

**Optimal Allocation (Part b):**

| Route               | Units | Cost | Total       |
|---------------------|-------|------|-------------|
| $A \rightarrow 3$   | 30    | 7    | 210         |
| $B \rightarrow 1$   | 40    | 3    | 120         |
| $B \rightarrow 2$   | 40    | 5    | 200         |
| $C \rightarrow 3$   | 40    | 4    | 160         |
| $D \rightarrow 1$   | 10    | 3    | 30          |
| $D \rightarrow 4$   | 80    | 4    | 320         |
| <b>Minimum Cost</b> |       |      | <b>1040</b> |

*Both methods converge to the same minimum transportation cost of **Rs. 1040**.  
Origin A holds 20 units at the dummy destination (excess supply, not shipped).*